| | | Target engagement | Dose | Efficacy | Safety |
|---|---|---|---|---|---|
| Autoimmune | **psoriasis**<br>anti-IL23A antibodies (3) | **2023** Acute local intradermal human IL-23 mouse skin assay dosed with mirikizumab… | **2018** Cynomolgus monkey repeat-dose toxicology (NOAEL) | **2019** Human psoriasis skin xenograft on AGR129 mice (anti-p19 clone 7G10) | **2018** Mouse intracellular-infection challenge with surrogate anti-p19 (BCG, Salmonella, Listeria) |
| | **ulcerative colitis**<br>S1PR1 agonists (3) | **2016** Rodent blood lymphocyte counts (ozanimod) | **2023** Rat and dog repeat-dose toxicology (etrasimod) | **2016** TNBS and CD4+CD45RBhi transfer colitis mice (ozanimod) | **2023** hERG current assay in vitro (etrasimod)<br>Ozanimod still slows the heart |
| Endocrine | **endometriosis**<br>GNRHR antagonists (3) | **2022** Rat HPG axis (LH in male rats; E2 in pro-oestrus female rats), linzagolix | **2022** Integrated PK-PD and systems-pharmacology model of linzagolix (E2, pelvic pain, bleedin… | **2024** Integrated PK-PD and systems-pharmacology model of linzagolix (E2, pelvic pain, bleedin… | **2022** Integrated PK-PD and systems-pharmacology model of…<br>Tox blind to BMD loss; liver held |
| | **secondary hyperparathyroidism**<br>CASR positive modulators (4) | **2015** 5/6-nephrectomy uremic rat | **2017** Population PK/PD model of etelcalcetide (PTH-calcium homeostasis, CaSR cooperativity) | **2017** 5/6-nephrectomy uremic rat | **2023** Rat gastric emptying and marmoset emesis vs cinacalcet |
| | **type 1 diabetes mellitus**<br>SLC5A2 inhibitors (4) | **2014** STZ-diabetic rat | **2008** STZ-diabetic rat | **2018** STZ-diabetic rat | **2018** Diet-induced obese rat, insulin-replete (dapagliflozin, 38-da…<br>DKA 3-4x placebo, unflagged by… |
| GI | **GERD**<br>ATP4A blockers (3) | **2019** Stimulated gastric acid secretion in dogs | **2017** Stimulated gastric acid secretion in dogs | **2019** Rat reflux oesophagitis vs esomeprazole | **2025** Rat repeat dosing: gastrin, emptying, microbiota |
| | **primary biliary cholangitis**<br>NR1H4 agonists (3) | **2024** Rodent FXR target-gene PD models | **2019** Rodent FXR target-gene PD models | **2023** Mdr2-/- (Abcb4-/-) mouse cholangiopathy | **2022** PXB chimeric humanized-liver mouse (FXR-agonist induction of…<br>IL-31 flat in PBC patients on cilofexor |
| Hematology | **hemophilia A**<br>F8 replacement / gene therapy (8) | **2025** Cynomolgus macaques with pre-existing anti-AAV5 antibodies given valoctocogene roxaparvovec… | **2017** F8-knockout mice + cynomolgus macaques | **2021** Hemophilia A dogs (Chapel Hill / Queen's colonies)<br>Mouse-stable BAX 888 lost in all… | **2023** Hemophilia A dogs (Chapel Hill / Queen's colonies) |
| | **hemophilia B**<br>F9 replacement / gene therapy (7) | **2017** Hemophilia B dogs, AAV8 canine FIX-Padua<br>FIX expression lost in 7 of 8 | **2017** Hemophilia B dogs, AAV8 canine FIX-Padua | **2024** Juvenile hemophilia B dogs, canine surrogate vector | **2024** Cynomolgus GLP tox, fidanacogene predecessor |
| | **PNH**<br>anti-C5 antibodies (4) | **2022** Hemolysis assay with C5 R885H variant | **2020** Cynomolgus PK/PD (crovalimab) | **2018** Complement hemolysis assay in human serum | **2020** Drug–target–drug complex assay, human C5 |
| Metabolic | **hyperlipidemia**<br>CETP inhibitors (3) | **2014** Human CETP/apoA-I transgenic mouse | **2014** CETP/apoA-I mouse; rat blood pressure<br>Exposure far below prediction | **2011** Human CETP loss-of-function genotypes<br>Outcome trial negative | **2014** Rat blood pressure + human adrenal H295R cells |
| | **hyperlipidemia**<br>HCAR2 agonists (3) | **2015** Rat and guinea pig plasma NEFA (GSK) | **2017** Rat and guinea pig plasma…<br>Exposure reached, no HDL-C response | **2017** Rat and guinea pig plasma…<br>GSK256073 phase 2a stopped for futility | **2017** Guinea pig ear-temperature flush model (GSK) |
| Oncology | **myelodysplastic syndrome**<br>HDAC (pan) inhibitors (3) | **2017** Mouse xenografts (HCT-116, later MOLM-13/SET-2 AML) | **2017** Mouse xenografts (HCT-116, later MOLM-13/SET-2 AML)<br>RP2D reached, combination failed | **2017** Mouse AML xenografts (MOLM-13, SET-2)<br>CR 18% vs 33% placebo | **2015** Dog telemetry and rat/dog repeat-dose toxicology (panobinostat IND package) |
| | **small cell lung carcinoma**<br>anti-PDCD1 antibodies (3) | **2023** In-vitro PD-1 receptor-occupancy titration of toripalimab on circulating human T… | **2020** Cynomolgus monkey 4-wee…<br>Target reached; CheckMate 331 missed OS | **2021** Tumour-mutational-burden patient-selection rule from…<br>CheckMate 331 missed OS;… | **2018** PD-1 knockout mice<br>Fatal pneumonitis in SCLC the… |
| Ophthalmology | **diabetic macular edema**<br>anti-VEGFA antibodies (3) | **2014** STZ-diabetic rat, Evans-blue blood-retinal-barrier leak (intravitreal VEGF Trap, VGT-NC-007) | **2023** Rabbit chronic retinal-neovascularisation leak model (aflibercept 8 mg) | **2021** STZ-diabetic rat, Evans-blue blood-retinal-barrier leak (intravitreal KH902/conbercept) | **2023** Cynomolgus repeat-dose…<br>PHOTON: no nasomucosal signal at 8 mg |
| | **dry eye syndrome**<br>anti-TNF antibodies (3) | **2023** Aged C57BL/6 mouse, spontaneous age-related dry eye (tanfanercept 0.25% eye drops, 8… | **2013** Rabbit topical ocular PK (ESBA105 eye drops) | **2023** Aged C57BL/6 mouse,…<br>Aged-mouse benefit, both Phase 3s negative | **2022** Rabbit topical ocular PK (systemic exposure margin) |
| Psy. | **insomnia**<br>HCRTR1/2 antagonists (6) | **2013** Rat, dog and rhesus sleep EEG/EMG (suvorexant) | **2022** PBPK-PD model of brain OX2 receptor occupancy (daridorexant, 65% occupancy threshold) | **2012** Rat, dog and rhesus sleep EEG/EMG (suvorexant) | **2019** Mouse emotion-induced cataplexy (lemborexant)<br>Animals no abuse; humans liked it |
| Respiratory | **cystic fibrosis**<br>CFTR correctors (10) | **2024** FRT cells expressing CFTR variants<br>Third corrector added nothing | **2015** Rat and dog toxicology (lumacaftor) | **2024** FRT cells expressing CFTR variants | **2025** Rat and dog tox (vanzacaftor) |
| | **cystic fibrosis**<br>CFTR potentiators (5) | **2026** Patient-derived primary nasal epithelial cells (HNE, Ussing chamber theratyping) | **2024** Patient-derived HBE cells (Ussing chamber) | **2024** FRT cells carrying each CFTR variant (Ussing chamber) | **2024** Rat toxicology (juvenile and adult, ivacaftor program) |
| Uro. | **interstitial cystitis**<br>anti-NGF antibodies (3) | **1997** Rodent chemical cystitis | **1997** Rodent chemical cystitis | **2022** ASP6294 vs anti-NGF IgG in arthritic rodents<br>ASP6294 missed IC pain endpoint | **2012** Primate and rodent joint tox<br>Joint harm missed; IC trials put on hold |
| Other | **atopic eczema**<br>HRH4 antagonists (3) | **2014** Mouse FITC Th2 dermatitis, H4R antagonists | **2021** Mouse FITC Th2 dermatitis,…<br>17-fold adriforant dose range, nothing worked | **2021** Dog acute atopic dermatitis (Df house-dust-mite-sensitised…<br>Adriforant worked in mice, not in… | **2014** Rat 6-month and monkey…<br>Agranulocytosis the tox species never showed |
| | **atopic eczema**<br>PDE4 inhibitors (5) | **2019** Stimulated human blood cells (PBMC cytokine release) | **2016** Minipig dermal toxicity (3- and 9-month) | **2023** Mouse chronic allergic contact dermatitis (difamilast)<br>Explant engaged; primary failed | **2016** Rabbit primary skin irritation |
| | **atopic eczema**<br>anti-IL33 antibodies (4) | **2024** IL-33-transgenic and IL-33/ST2-knockout mice (MC903 dermatitis) | **2024** IL-33-transgenic and…<br>Target saturated at 300 mg; still failed | **2024** IL-33-transgenic and IL-33/ST2-knockout mice (MC903…<br>Four antibodies, four phase 2 misses | **2024** Cynomolgus PK and IL-1-family specificity panel (torudokimab) |